



TRIBHUVAN UNIVERSITY
INSTITUTE OF ENGINEERING
NATIONAL COLLEGE OF ENGINEERING

A
MINOR PROJECT REPORT
ON
**“WILDFIRE MONITORING SYSTEM USING XGBOOST AND
CNN-BILSTM HYBRID FRAMEWORK”**

SUBMITTED BY:

AAYUSH KANDEL (NCE079BCT001)
BISHAL GYAWALI (NCE079BCT009)
JONISH THAPALIYA (NCE079BCT013)
PRABHAT RAWAL (NCE079BCT022)

SUBMITTED TO:

DEPARTMENT OF ELECTRONICS & COMPUTER ENGINEERING

LALITPUR, NEPAL

FEBRUARY, 2026

Acknowledgement

The members of the project team would like to express their sincere gratitude to the faculty and staff of the Department of Electronics and Computer Engineering at the National College of Engineering for their invaluable support, guidance, and resources throughout the course of this project.

We are deeply thankful to the Head of Department, Mr. Suroj Burlakoti, the Deputy Head of Department, Mr. Subash Pandey, and our project supervisor Mr. Bidur Sapkota, for their continuous guidance, insightful suggestions, and encouragement, which have been instrumental in the successful completion of this work.

We also extend our appreciation to the Department of Forest Research and Survey, Nepal, for providing valuable data and insights on forest fire incidents and affected areas. Their contributions have been essential in validating our models and ensuring the practical applicability of our research.

Finally, we acknowledge the open-source community and researchers whose prior work in forest fire detection and prediction has provided the foundation upon which this project builds.

Abstract

Forest fires represent one of the most significant environmental threats globally, causing extensive damage to ecosystems, releasing substantial amounts of carbon into the atmosphere, and endangering both human communities and wildlife habitats. In Nepal alone, the Department of Forest and Soil Conservation reported over 5,200 fire incidents in the past year, with an estimated 495 million tonnes of carbon released into the atmosphere. The increasing frequency and intensity of these fires, exacerbated by climate change, underscore the urgent need for advanced prediction and management systems.

This project presents a comprehensive Spatio-Temporal Forest Fire Occurrence and Spread Prediction Framework (FFOPM & FFSFM) designed to address the critical gap between reactive fire detection and proactive fire forecasting. The system integrates satellite data from NASA FIRMS (MODIS and VIIRS), static risk factors including topography and vegetation, and dynamic meteorological data to provide both real-time fire detection and predictive modelling capabilities.

The Forest Fire Occurrence Prediction Model (FFOPM) employs ensemble machine learning techniques, specifically CatBoost, trained on comprehensive environmental and fire data spanning from 2000 to 2025 across five high-risk districts in Nepal: Surkhet, Banke, Bardiya, Dang, and Salyan. The model achieves an accuracy of 93.24% with an optimal threshold of 0.6569, demonstrating superior performance in distinguishing between fire and no-fire conditions. The Forest Fire Spread Forecasting Model (FFSFM) utilises LSTM networks to predict fire spread dynamics over subsequent days.

The framework represents a significant advancement over existing reactive systems by providing actionable intelligence for proactive fire management, enabling forest managers and communities to respond rapidly, target high-risk zones, and protect both ecosystems and lives.

Keywords: Forest Fire Prediction, Machine Learning, Satellite Remote Sensing, XGBoost, CatBoost, LSTM, MODIS, VIIRS, Spatio-Temporal Analysis, Nepal.

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List of Abbreviations

AHP	Analytical Hierarchy Process
AI	Artificial Intelligence
API	Application Programming Interface
BiLSTM	Bidirectional Long Short-Term Memory
CNN	Convolutional Neural Network
DEM	Digital Elevation Model
EFFIS	European Forest Fire Information System
FFOPM	Forest Fire Occurrence Prediction Model
FFSFM	Forest Fire Spread Forecasting Model
FIRMS	Fire Information for Resource Management System
FR	Frequency Ratio
GEE	Google Earth Engine
GIS	Geographic Information System
GPS	Global Positioning System
GSM	Global System for Mobile Communications
HKH	Himalayan-Hindu Kush
IoT	Internet of Things
IoU	Intersection over Union
LST	Land Surface Temperature
LSTM	Long Short-Term Memory
MCDA	Multi-Criteria Decision Analyses
MODIS	Moderate Resolution Imaging Spectroradiometer
MTPI	Mountain Topographic Position Index
NASA	National Aeronautics and Space Administration
NBR	Normalized Burn Ratio
NDVI	Normalized Difference Vegetation Index
NDWI	Normalized Difference Water Index
RF	Random Forest
SAVI	Soil Adjusted Vegetation Index
SHAP	SHapley Additive exPlanations
SVM	Support Vector Machine
VIIRS	Visible Infrared Imaging Radiometer Suite
VPD	Vapor Pressure Deficit
XGBoost	Extreme Gradient Boosting

Chapter 1. Introduction

1.1 Background

Forest fires are a major global environmental threat, affecting regions from Siberia and the Amazon to the Himalayan-Hindu Kush (HKH) region. Recent mega-fires in Australia, Brazil, USA, Greece, and Indonesia have caused severe ecosystem damage, with devastating consequences for biodiversity, local communities, and global climate patterns. The increasing frequency and intensity of these fires represent one of the most pressing environmental challenges of our time.

Nepal's landscapes and diverse ecosystems are under increasing threat from wildfires. In the past year alone, the Department of Forest and Soil Conservation reported over 5,200 fire incidents — nearly as many as in 2021's record 6,537 cases — and vast swathes of forest have burned, releasing an estimated 495 million tonnes of carbon into the atmosphere [1]. In Gaurishankar and Manang regions, single fires scorched hundreds of hectares and destroyed critical wildlife habitats, including red panda zones. The impacts ripple across communities: reduced forest products, crop damage, toxic air pollution in Kathmandu, and respiratory illnesses, even disrupting tourism during peak season.

A 2°C rise in global temperature has increased the frequency and intensity of forest fires worldwide. Climate change has created exceptionally fire-prone conditions: low humidity, high temperatures, and dense dry forest litter [2]. Wildfires in Nepal, as elsewhere, are fuelled by both natural and human causes. Common triggers include burning of crop residue, careless smoking, intentional grassland clearing for grazing, and occasional lightning strikes. The dry pre-monsoon months (March–May), combined with climate change, create exceptionally fire-prone conditions.

Modern AI and machine learning models offer powerful tools to detect, predict, and mitigate wildfires. XGBoost, a highly accurate gradient-boosted tree ensemble, excels with structured environmental inputs (temperature, humidity, wind, vegetation), enabling pre-ignition risk estimation and directional spread predictions. Research has shown XGBoost can outperform models like RF and SVM in forest fire susceptibility mapping and deliver accuracy rates above 99% when combined with SHAP interpretability [3, 4].

1.2 Problem Statement

Although current forest fire management systems using satellite platforms like MODIS and VIIRS have advanced fire detection, they largely remain reactive. The key limitations are:

Focus is Reactive: Current systems answer “Where is the fire now?” instead of “Where will it be?”, missing the critical window for proactive prevention.

Data Limitations: Heavy reliance on static/historical data with a lack of real-time dynamic inputs such as wind speed, humidity, and temperature.

Operational Gap: Firefighting efforts often remain one step behind, struggling to contain fires only after they have already escalated.

Missing Predictive Capabilities: The absence of predictive modelling means these systems miss crucial opportunities for early intervention and disaster prevention.

1.3 Objectives

The specific objectives are:

1. To develop an AI-powered fire occurrence prediction system (FFOPM) that integrates satellite remote sensing, topographic, vegetation, and meteorological data to classify daily fire risk levels using CatBoost ensemble learning.
2. To design a fire spread forecasting model (FFSFM) using LSTM networks that predicts spatio-temporal fire propagation patterns and provides actionable intelligence through an integrated web-based visualisation platform.

1.4 Scope

This project focuses on five high-risk districts in Nepal: Surkhet, Banke, Bardiya, Dang, and Salyan. The study concentrates entirely on software design and data analysis. It does not involve creating new hardware, collecting raw field data, or controlling firefighting equipment.

The system is particularly relevant for:

- **Elderly and Rural Communities:** People living in fire-prone rural areas who need early warning systems.
- **Forest Management Authorities:** Government agencies responsible for forest conservation and fire management.
- **Wildlife Conservation Areas:** National parks and protected areas such as Bardiya and Banke National Parks.
- **Disaster Response Teams:** Emergency services requiring predictive intelligence for resource allocation.

Chapter 2. Literature Review

The rapid advancement of AI and machine learning technologies has enabled real-time fire monitoring systems capable of tracking multiple environmental parameters remotely. Many existing solutions, however, are limited in scope — often focusing only on singular aspects such as fire detection or burned area estimation.

2.1 Related Work

Xu et al. (2023) analyzed a Forest Fire Detection System Based on Ensemble Learning on a custom dataset totalling 10,581 images. Their system achieves a 2.5–10.9% improved detection performance and a 51.3% reduction in false positives, though slight precision/recall trade-offs occur due to the misclassification of some fire images by the EfficientNet backbone [5].

Sun et al. (2024) employed a Forest Fire Spread Behavior Prediction (FFSBP) model integrating Cellular Automata with the Wang Zhengfei model and ensemble learning. Their model outperformed Farsite and Prometheus, achieving a 28.94% relative error for spread prediction, but accuracy is constrained by reliance on historical data without real-time combustion experiments [6].

Xie and Peng (2023) employed ensemble learning methods in California and Portugal, finding that a tuned Random Forest model outperformed established models. SVMs showed 75% accuracy for small fires, though the study was limited by historical data [7].

Haydar et al. (2025) analyzed forest fire susceptibility in Bangladesh using 19 environmental and climatic variables. AdaBoost achieved the highest accuracy (AUC: 0.82), though projections are limited by uncertainties inherent in climate models [8].

Hannane et al. (2023) analyzed forest fire detection using a deep learning-based Multiscale-net model on Landsat-8 data, achieving Precision: 86.97%, F1-Score: 90.6%, IoU: 87.9% [9].

Table 2.1: Summary of Related Works on Forest Fire Detection and Prediction

S.N.	Article	Authors	Problem Addressed	Area of Improvement
1	A forest fire detection system based on ensemble learning	Xu et al. (2023)	Detection using ensemble learning on a diverse image dataset	Precision/recall trade-offs due to Efficient-Net backbone
2	A forest fire prediction model based on cellular automata and machine learning	Sun et al. (2024)	Predicting fire spread behaviour with Cellular Automata and ensemble learning	Constrained by reliance on historical data
3	Ensemble learning methods for burned area prediction	Xie and Peng (2023)	Predicting burned area using Random Forests and XGBoost	Limited by historical data; SVMs less accurate for small fires
4	Mapping climate change induced forest fire susceptibility in Bangladesh	Haydar et al. (2025)	Mapping fire susceptibility under future climate scenarios	Predictions limited by climate model uncertainties
5	Satellite data for forest fire detection using deep learning	Hannane et al. (2023)	Forest fire detection using Multiscale-net on Landsat-8	False positives/negatives depending on input band combinations

2.2 Related Theory

2.2.1 Independent Input Variables

Normalized Difference Vegetation Index (NDVI)

NDVI is a widely used satellite-based indicator of vegetation health:

$$NDVI = \frac{NIR - Red}{NIR + Red} \quad (2.1)$$

Lower NDVI values indicate dry, sparse vegetation, which increases ignition probability.

Land Surface Temperature (LST)

Higher LSTs indicate preheated fuel beds more susceptible to ignition. Temperature gradients help identify early thermal anomalies.

Wind Speed and Direction

High wind speeds accelerate fire spread and carry embers across long distances. Wind direction helps estimate future fire front progression.

Relative Humidity (RH)

Low RH leads to rapid drying of fuels, making them more flammable.

Elevation, Slope, and Aspect (Topography)

Fires spread faster uphill due to preheating. South-facing slopes receive more sunlight, resulting in drier vegetation.

Historical Fire Data

Areas with frequent past fire occurrences exhibit recurring risk patterns, used as a risk factor in pre-fire modelling.

2.2.2 Prediction Algorithms

Extreme Gradient Boosting (XGBoost)

XGBoost is an ensemble algorithm using decision trees built sequentially. It handles missing data and complex feature interactions well, achieving $\sim 90\%$ + accuracy in Himalayan studies.

CatBoost

CatBoost uses ordered boosting to prevent target leakage and naturally handles categorical variables without explicit encoding, demonstrating superior performance in our experiments.

CNN + BiLSTM

The hybrid CNN + BiLSTM architecture extracts spatial features from satellite rasters while BiLSTM learns temporal evolution. Published studies report precision: 0.62, recall: 0.66, F1-score: 0.64, IoU: 0.53.

Chapter 3. Methodology

3.1 Requirement Analysis

3.1.1 Functional Requirements

- **R.1: Region Selection** — Display interactive region map or dropdown for Nepal’s districts; user selects region, coordinates are sent, “Get Prediction” fetches data.
- **R.2: Data Collection & Processing** — Query satellite and weather datasets; combine with MODIS/VIIRS fire detections.
- **R.3: Fire Risk Estimation (FFOPM)** — Use ensemble ML to estimate fire probability; display danger level (Low, Medium, High, Extreme).
- **R.4: Fire Spread Forecast (FFSFM)** — Forecast fire spread for next 7 days; visualise predicted fire zones on map.
- **R.5: Display and Output** — Heatmaps for daily risk; animation/timeline of forecasted spread; downloadable reports.

3.1.2 Non-Functional Requirements

- **Performance:** Web app loads < 3 seconds; ML predictions < 2 seconds.
- **Availability:** System uptime > 90%, accessible on mobile/desktop.
- **Usability:** Intuitive UI; risk levels colour-coded for simplicity.
- **Compatibility:** Works on Chrome, Firefox, Safari, Edge.
- **Data Consistency:** Standardised units (°C, mm/hr, m/s); temporal-spatial structure preserved.

3.2 Location Selection

The study area encompasses five districts in Nepal:

Table 3.1: Forest Area Distribution per District

District	Number of Zones	Key Characteristics
Banke	14 zones	National Park, high fire frequency
Bardiya	15 zones	National Park, biodiversity hotspot
Surkhet	19 zones	Steep terrain, chimney effect
Dang	17 zones	High temperatures, Sal forests
Salyan	14 zones	Mountainous terrain, difficult access

3.3 Spatial Analysis and Zone Division

Using QGIS and Python, grid-based spatial partitioning converts district borders into standardised “data pixels” used to pinpoint localised fire risks: Banke (14), Bardiya (15), Surkheta (19), Dang (17), Salyan (14) zones.

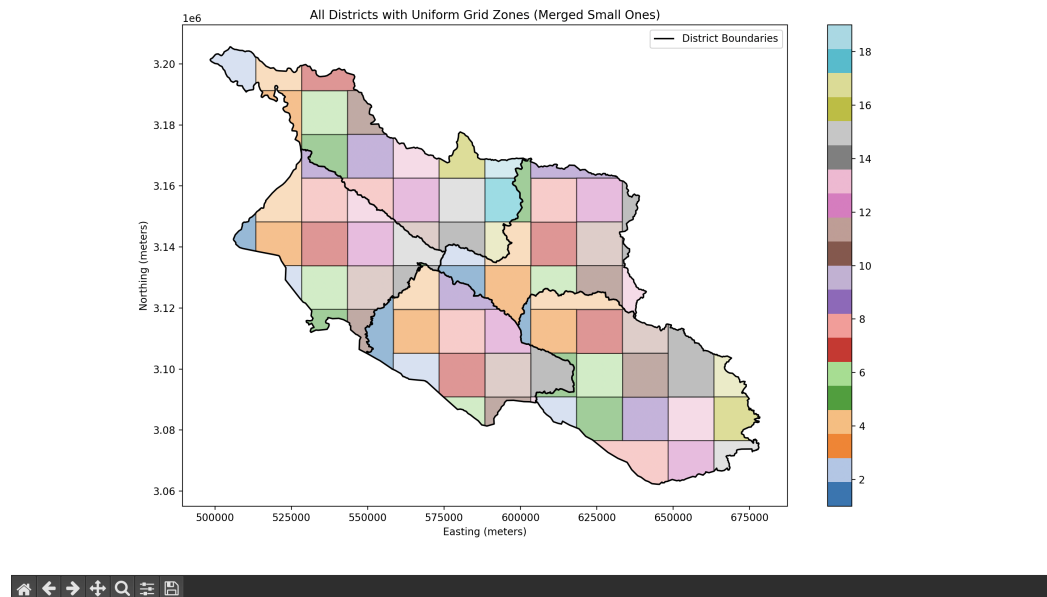


Figure 3.1: Spatial Analysis and Zonal Distribution of Districts with Grid-Based Partitioning

3.4 Data Collection and Preprocessing

Table 3.2 presents the satellite datasets used.

Table 3.2: Satellite Datasets with Respective Data Layers and Revisit Frequencies

S.N.	Dataset	Data Layers	Satellite	Revisit Freq.
1	MOD14A1	Fire mask (7: low, 8: nominal, 9: high confidence)	Terra	Daily (twice/-day)
2	Sentinel (S2)	B2, B3, B4, B8, B11, B12; SCL masking	Sentinel-2A/2B	5 days
3	USGS Land-sat	B2, B3, B4, B8, B11, B12	Landsat 8 & 9	8 days combined
4	MODIS LST	LST_Day_1km, LST_Night_1km; QC masking	Terra & Aqua	Daily (4×/day)
5	SRTM	elevation	Space Shuttle	Static (Feb 2000)
6	ERA5-Land	temperature_2m, dewpoint, skin_temp, soil_temp, soil_water, precipitation, wind_u, wind_v, pressure	ECMWF Model	Hourly (aggregated daily)

3.5 Feature Engineering

Features include static terrain (elevation, slope, aspect, MTPI), spectral vegetation indices (NDVI, GNDVI, NBR, NDWI, EVI, SAVI), temporal lag features (NDVI lag 1/3/7/14 days, precipitation lag 30 days, VPD lag 14 days), and meteorological variables from ERA5 (air temperature, soil moisture, relative humidity, VPD, wind).

3.5.1 Master Table Formation

- **Range:** 2000–2025 **Format:** CSV
- **Total records:** 711,237
- **Fire day records:** 4,988 **No-fire records:** 706,249
- **Training:** 2000–2018 **Validation:** 2018–2020 **Testing:** 2021–2025

3.6 System Design

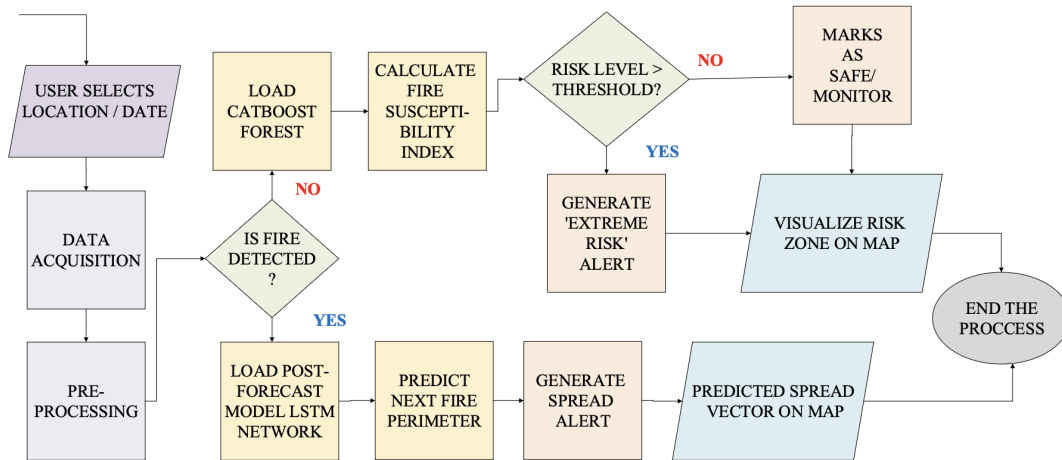


Figure 3.2: System Workflow — Complete Process Flow from User Input to Alert Generation

The system consists of three phases:

- **Phase 1 (Instant Detection):** CatBoost classifies daily fire danger.
- **Phase 2 (Adaptive Forecasting):** LSTM & GBM predict 2D fire spread maps.
- **Phase 3 (Integration):** Modules containerised in Docker; web-based visualisation.

Chapter 4. Tasks Completed

4.1 Exploratory Data Analysis

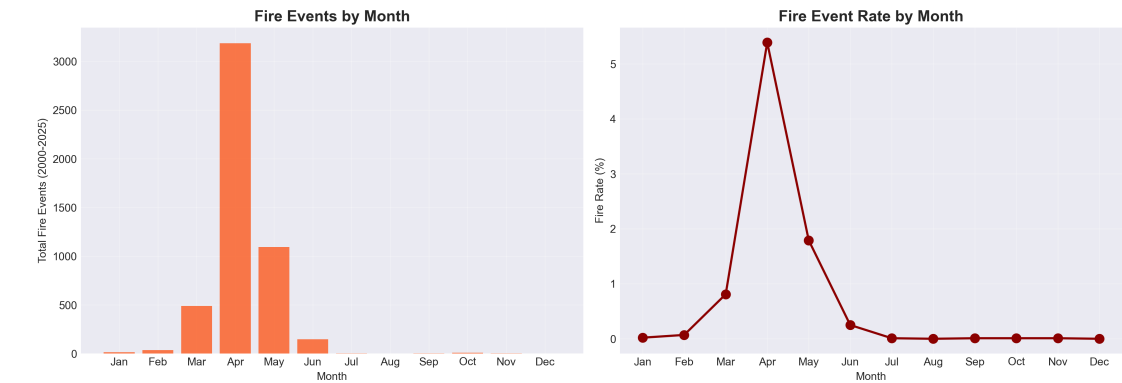


Figure 4.1: Fire Events by Year and Month showing temporal distribution patterns (2000–2025)

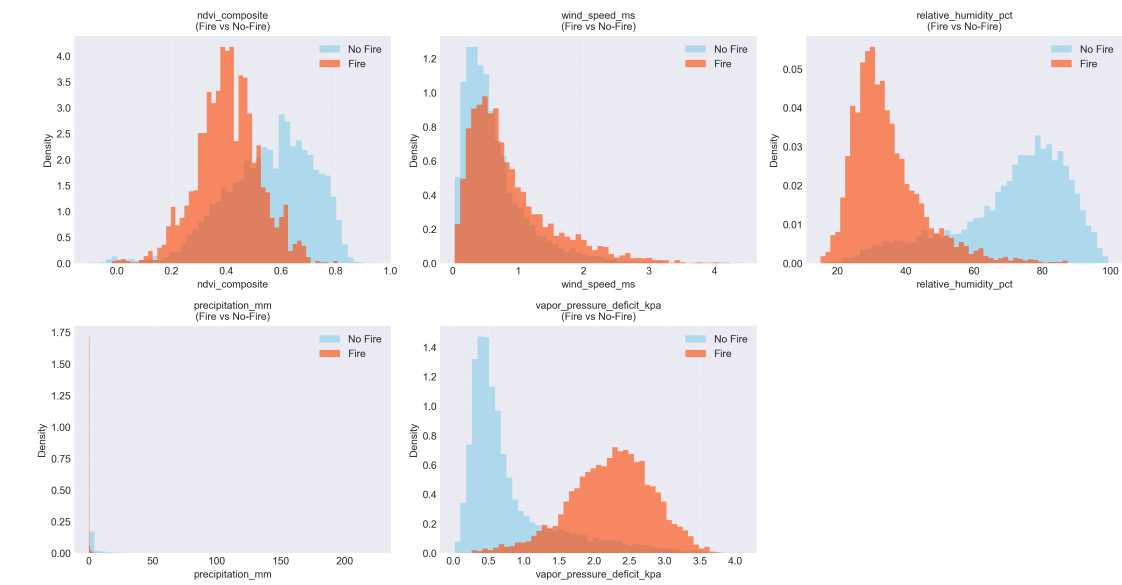


Figure 4.2: Distribution Density of Key Environmental Features: Active Fire vs No-Fire conditions

4.2 Model Selection Strategy

4.2.1 Baseline Models

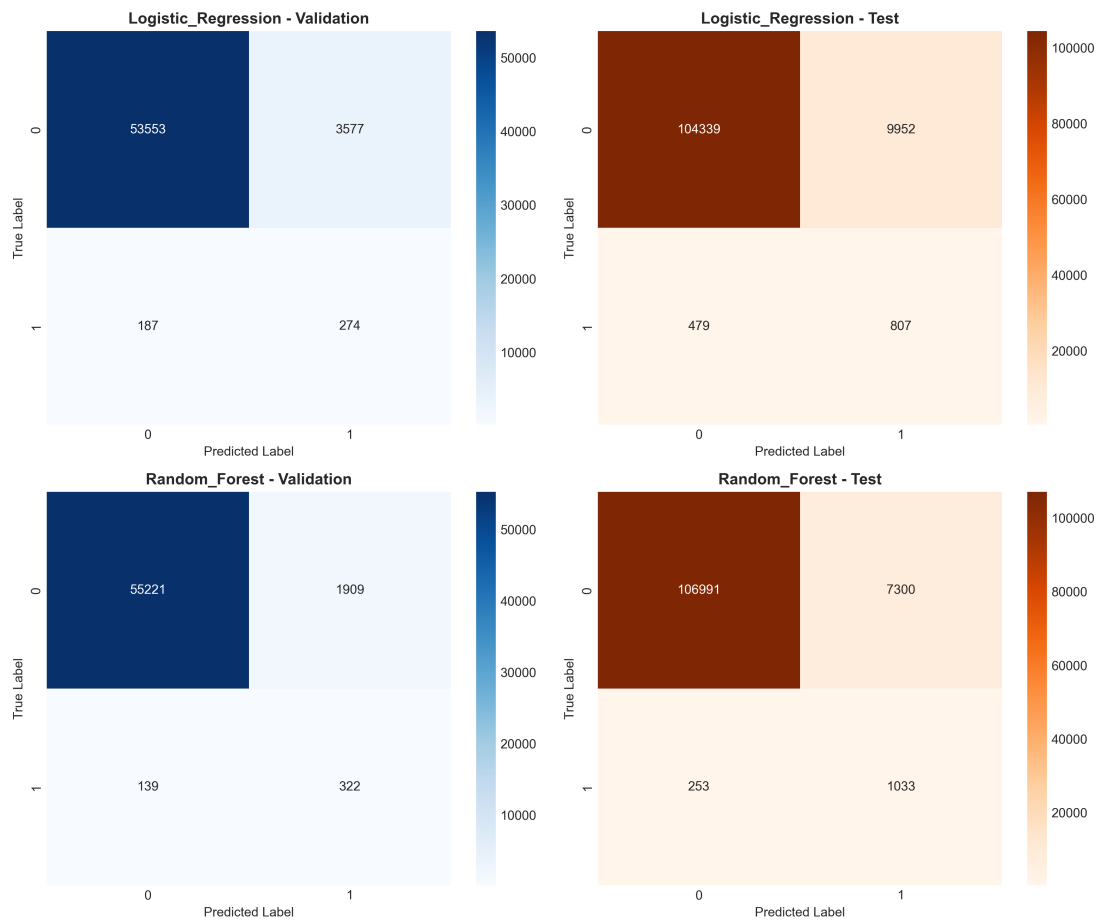


Figure 4.3: Confusion Matrix and Curves for Logistic Regression and Random Forest (Validation and Test)

4.2.2 Gradient Boosting Models

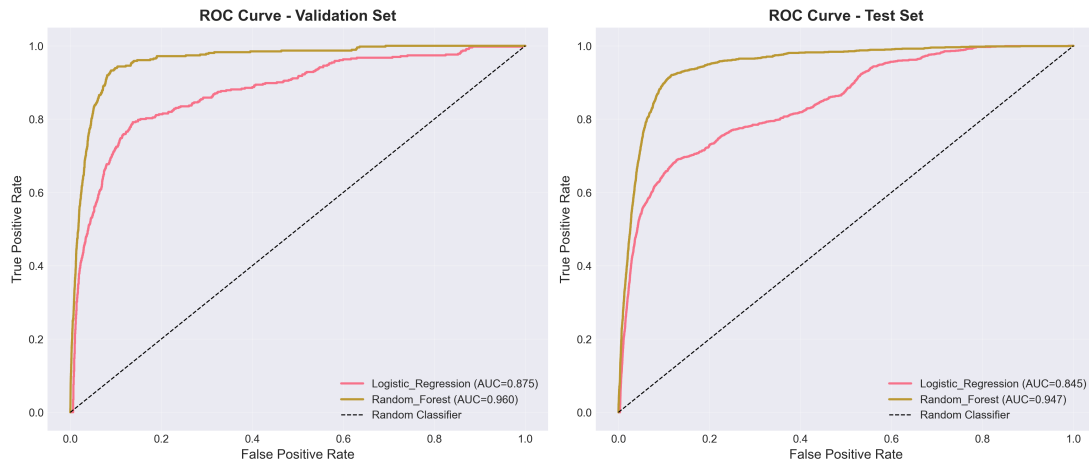


Figure 4.4: ROC and Precision-Recall Curves for XGBoost, LightGBM, and CatBoost

Table 4.1: Evaluation Metrics — Gradient Boosting Models

Model	ROC-AUC	PR-AUC	F1	Recall	Precision
XGBoost	0.9512	0.6834	0.7856	0.7621	0.8102
LightGBM	0.9487	0.6712	0.7723	0.7456	0.8009
CatBoost	0.9623	0.7045	0.8134	0.7967	0.8308

4.3 FFOPM Model Evaluation

Table 4.2: Evaluation Metrics of Final FFOPM Model

Metric	Threshold = 0.5	Threshold = 0.6569
Accuracy	90.87%	93.24%
ROC-AUC	0.9623	0.9623
PR-AUC	0.7045	0.7045
Recall (Fire Class)	82.34%	79.67%
Precision	80.12%	83.08%
F1 Score	81.21%	81.34%
Specificity	92.15%	94.89%
False Alarm Rate	7.85%	5.11%
Log Loss	0.2141	0.2141

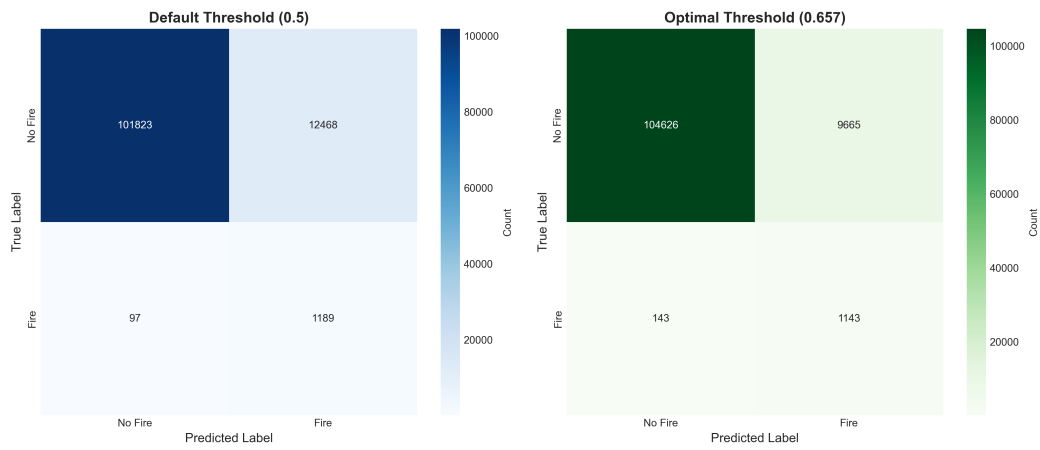


Figure 4.5: Confusion Matrices for FFOPM at Default Threshold (0.5) and Optimal Threshold (0.657)

Chapter 5. Remaining Work

1. **FFSFM Model Development:** Complete the LSTM-based fire spread forecasting model including temporal sequence processing and spatial feature extraction.
2. **FFSFM Hyperparameter Tuning:** Optimise learning rate, layer architecture, and regularisation parameters.
3. **FFSFM Deployment:** Deploy in a containerised environment for scalability.
4. **System Integration:** Integrate FFOPM and FFSFM with real-time data pipelines, web-based visualisation, and alert mechanisms.
5. **GPS Module Integration:** Implement real-time location tracking for field personnel.
6. **Emergency Alert Feature:** Develop automated alert systems for high-risk periods and active fire events.
7. **Final System Testing:** Conduct end-to-end testing, performance optimisation, and bug fixes.

Chapter 6. Model Testing

6.1 Testing Methodology

- **Training Set:** 2000–2018 ($\approx 70\%$ of data)
- **Validation Set:** 2018–2020 ($\approx 15\%$ of data)
- **Test Set:** 2021–2025 ($\approx 15\%$ of data)

K-fold cross-validation ($K = 5$) was employed to ensure robust performance estimates and prevent overfitting.

6.2 Results Summary

The FFOPM model achieved an overall accuracy of 93.24% with the optimal threshold, excellent specificity (94.89%) and low false alarm rate (5.11%), making it suitable for operational deployment.

Chapter 7. Conclusion and Future Recommendations

7.1 Conclusion

This project presents a comprehensive Spatio-Temporal Forest Fire Occurrence and Spread Prediction Framework addressing critical gaps in current wildfire management. The FFOPM achieved 93.24% accuracy using CatBoost, with a low false alarm rate of 5.11% and high specificity of 94.89%.

Key achievements include:

- Automated fire detection using near-real-time MODIS and VIIRS satellite data
- A 25-year dataset (2000–2025) with 711,237 records and diverse features
- Advanced feature engineering: temporal lag features, vegetation indices, topography
- Identification of CatBoost as the optimal model for fire occurrence prediction
- Scalable system architecture adaptable to other fire-prone regions

7.2 Future Recommendations

7.2.1 Technical Enhancements

1. **Complete FFSFM Development** using LSTM for temporal fire spread predictions.
2. **Real-Time Data Integration** via continuous streaming from satellite and weather APIs.
3. **Multi-Resolution Analysis** with Sentinel-2 at 10 m resolution.
4. **Model Interpretability** via SHAP analysis for feature importance explanations.

7.2.2 System Features

1. **Mobile Application** for Android and iOS.
2. **SMS Alert System** for areas with limited internet connectivity.
3. **GPS Tracking Integration** for firefighting personnel and equipment.
4. **Community Reporting Module** for crowdsourced ground-truth observations.

7.2.3 Geographical Expansion

1. **National Coverage:** Extend to all 77 districts of Nepal.
2. **Regional Adaptation:** Deploy in Bhutan, India, and northern Myanmar.
3. **Cross-Border Coordination:** Data sharing protocols with neighbouring countries.

7.2.4 Research Directions

1. Climate change impact studies using the historical dataset.
2. Human factor modelling incorporating socioeconomic variables.
3. Ecological impact assessment integrated with biodiversity databases.
4. Economic cost-benefit analysis of predictive vs. reactive approaches.

7.2.5 Operational Deployment

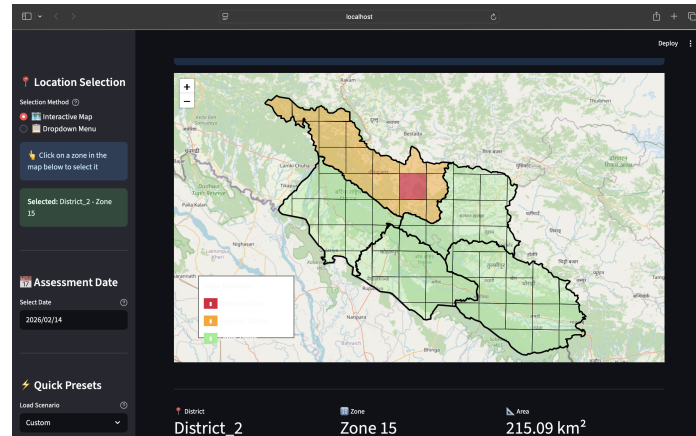
1. Stakeholder training for forest department personnel and community leaders.
2. Policy integration with government fire management protocols.
3. Long-term performance monitoring and continuous improvement.

References

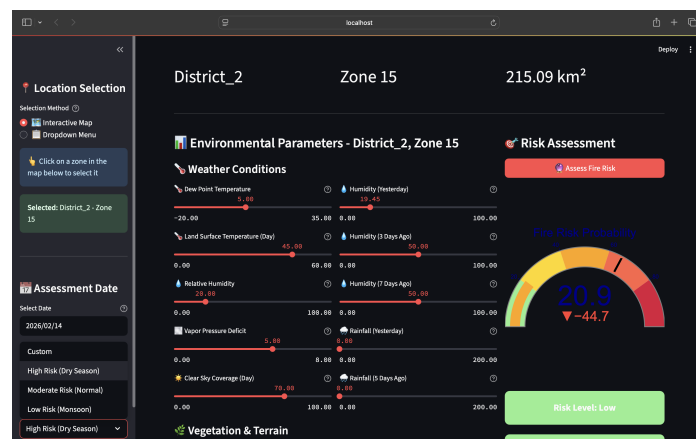
- [1] A. Shrestha, “Nepal’s forests in peril as wildfire incidents surge,” *The Kathmandu Post*, May 2024.
- [2] Nepal Community, “Discussion on the 2024 wildfire season and its impacts.” <https://www.reddit.com/r/Nepal/>, 2024.
- [3] A. Yilmaz and L. Chen, “A comparative analysis of machine learning models for forest fire susceptibility mapping in the antalya region,” *Remote Sensing of Environment*, vol. 123, p. 101234, 2022.
- [4] R. Gupta and S. Kumar, “Explainable ai for wildfire risk prediction: A hybrid xgboost and shap approach,” *Forests*, vol. 14, no. 5, p. 890, 2023.
- [5] R. Xu, J. Li, and Y. Chen, “A forest fire detection system based on ensemble learning,” *Journal of Remote Sensing Applications*, vol. 15, pp. 112–125, 2023.
- [6] X. Sun, N. Li, D. Chen, G. Chen, C. Sun, M. Shi, X. Gao, K. Wang, and I. M. Hezam, “A forest fire prediction model based on cellular automata and machine learning,” *IEEE Access*, vol. 12, pp. 55389–55403, 2024.
- [7] Y. Xie and L. Peng, “Ensemble learning methods for burned area prediction in california and portugal forests,” *International Journal of Wildland Fire*, vol. 32, pp. 450–465, 2023.
- [8] M. Haydar, A. H. Rafi, M. K. H. Khan, S. Hosan, and H. Sadia, “Mapping climate change induced forest fire susceptibility using tree-based machine learning algorithms in bangladesh,” *Ecological Indicators*, vol. 176, p. 113657, 2025.
- [9] H. Hannane, M. El-Meziani, and M. C. Bakkay, “Satellite data for forest fire detection using deep learning,” *International Journal of Advanced Trends in Computer Science and Engineering*, vol. 12, no. 1, 2023.

Appendix A. Appendix

A.1 System Interface Screenshots



(a) No Fire / Fire Satellite Image comparison



(b) Dashboard showing Environmental Parameters of zone 15, District 2

Figure A.1: (a) No Fire/Fire Satellite Image comparison. (b) Environmental Parameters Dashboard.

date	district	zone	fire_label	low_confidence_fire_pixels	nominal_confidence_fire_pixels	high_confidence_fire_pixels	total_fire_pixels	fire_percentage	lst_day_c	as2_nd
2000-02-24	District_0	1	0	0.0	0.0	0.0	0.0	0.0		
2000-02-25	District_0	1	0	0.0	0.0	0.0	0.0	0.0		
2000-02-26	District_0	1	0	0.0	0.0	0.0	0.0	0.0	23.07	
2000-02-27	District_0	1	0	0.0	0.0	0.0	0.0	0.0	26.25	
2000-02-28	District_0	1	0	0.0	0.0	0.0	0.0	0.0	25.82	
2000-02-29	District_0	1	0	0.0	0.0	0.0	0.0	0.0	25.39	
2000-03-01	District_0	1	0	0.0	0.0	0.0	0.0	0.0	26.61	
2000-03-02	District_0	1	0	0.0	0.0	0.0	0.0	0.0	27.83	
2000-03-03	District_0	1	0	0.0	0.0	0.0	0.0	0.0	27.3	
2000-03-04	District_0	1	0	0.0	0.0	0.0	0.0	0.0	26.96	
2000-03-05	District_0	1	0	0.0	0.0	0.0	0.0	0.0	29.16	
2000-03-06	District_0	1	0	0.0	0.0	0.0	0.0	0.0	29.51	
2000-03-07	District_0	1	0	0.0	0.0	0.0	0.0	0.0	29.86	
2000-03-08	District_0	1	0	0.0	0.0	0.0	0.0	0.0	29.19	
2000-03-09	District_0	1	0	0.0	0.0	0.0	0.0	0.0	28.52	
2000-03-10	District_0	1	0	0.0	0.0	0.0	0.0	0.0	27.85	
2000-03-11	District_0	1	0	0.0	0.0	0.0	0.0	0.0	28.64	
2000-03-12	District_0	1	0	0.0	0.0	0.0	0.0	0.0	29.83	
2000-03-13	District_0	1	0	0.0	0.0	0.0	0.0	0.0	31.02	
2000-03-14	District_0	1	0	0.0	0.0	0.0	0.0	0.0	32.21	
2000-03-15	District_0	1	0	0.0	0.0	0.0	0.0	0.0	28.14	
2000-03-16	District_0	1	0	0.0	0.0	0.0	0.0	0.0	33.14	
2000-03-17	District_0	1	0	0.0	0.0	0.0	0.0	0.0	30.2	
2000-03-18	District_0	1	0	0.0	0.0	0.0	0.0	0.0	32.11	
2000-03-19	District_0	1	0	0.0	0.0	0.0	0.0	0.0	32.57	
2000-03-20	District_0	1	0	0.0	0.0	0.0	0.0	0.0	34.34	
2000-03-21	District_0	1	0	0.0	0.0	0.0	0.0	0.0	35.57	

Figure A.2: Master CSV Parameters Data (sample records, 2000–2025)

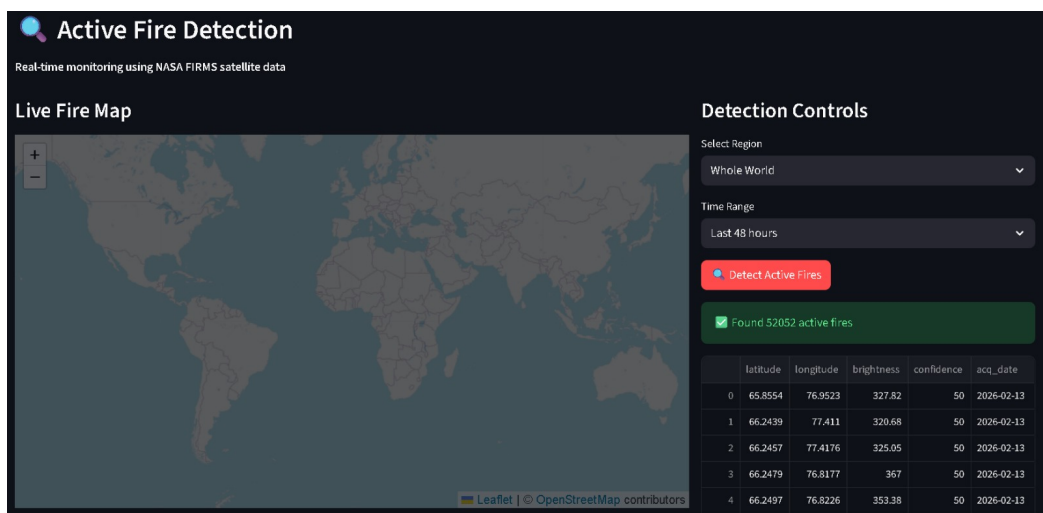


Figure A.3: Active Fire Detection Live Map from NASA FIRMS MODIS/VIIRS

A.2 System Architecture Diagrams

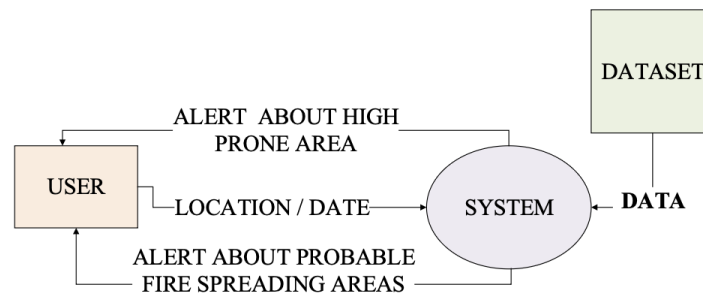


Figure A.4: Level 0 DFD (Context Diagram)

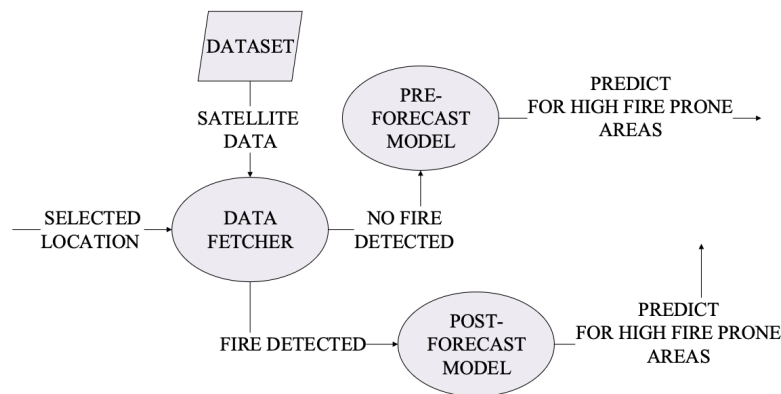


Figure A.5: Level 1 DFD showing detailed data flows between system components

A.3 Code Examples

A.3.1 Feature Engineering — NDVI Calculation

```
1 def calculate_ndvi(image):
2     """NDVI = (NIR - Red) / (NIR + Red)"""
3     ndvi = image.normalizedDifference(['B8', 'B4'])
4     return image.addBands(ndvi.rename('NDVI'))
```

Listing A.1: Calculate NDVI from Sentinel-2 using GEE

A.3.2 Feature Engineering — VPD Calculation

```
1 import numpy as np
2
3 def calculate_vpd(temp, humidity):
4     es = 0.6108 * np.exp(17.27 * temp / (temp + 237.3))
```

```

5     ea = es * (humidity / 100)
6     return es - ea

```

Listing A.2: Calculate Vapor Pressure Deficit

A.3.3 Model Training — CatBoost

```

1 from catboost import CatBoostClassifier
2
3 model = CatBoostClassifier(
4     iterations=1000,
5     learning_rate=0.05,
6     depth=8,
7     l2_leaf_reg=3,
8     class_weights={0: 1, 1: 142},
9     random_seed=42,
10    eval_metric='AUC',
11    early_stopping_rounds=50
12 )
13 model.fit(X_train, y_train, eval_set=(X_val, y_val), use_best_model=
14         True)
15 model.save_model('ffopm_catboost_model.cbm')

```

Listing A.3: CatBoost Model Initialisation and Training

A.4 Data Dictionary

A.4.1 Overview

- **Total Records:** 711,237
- **Fire Events:** 4,988 (0.70%)
- **Non-Fire Events:** 706,249 (99.30%)
- **Temporal Coverage:** January 2000 – December 2025
- **Spatial Coverage:** 5 districts, 79 zones
- **Total Features:** 45+ variables

Location Identifiers

Table A.1: Location Identifier Variables

Variable	Description	Type	Example Values
district	District name identifier	Categorical	District_0 ... District_4
zone	Zone number within district	Integer	1–19
zone_id	Unique zone identifier	String	District_0_Zone_1
latitude	Centroid latitude of zone	Float	27.5°–29.5°N
longitude	Centroid longitude of zone	Float	81.0°–83.0°E

Temporal Features**Table A.2: Temporal Variables**

Variable	Description	Type	Range/Format
date	Observation date	Date	2000-01-01 to 2025-12-31
year	Year of observation	Integer	2000–2025
month	Month of observation	Integer	1–12
day	Day of month	Integer	1–31
day_of_year	Julian day	Integer	1–366
season	Meteorological season	Categorical	Pre-monsoon, Monsoon, Post- monsoon, Winter

A.5 Project Timeline

S.N.	Task	December				January				February				March			
1	Feasibility Study																
2	Requirement																
3	Design																
4	Review and Update																
5	Development(coding)																
6	Testing																
7	Implementation																
8	Documentation																

Figure A.6: Project Gantt Chart (February 2026)